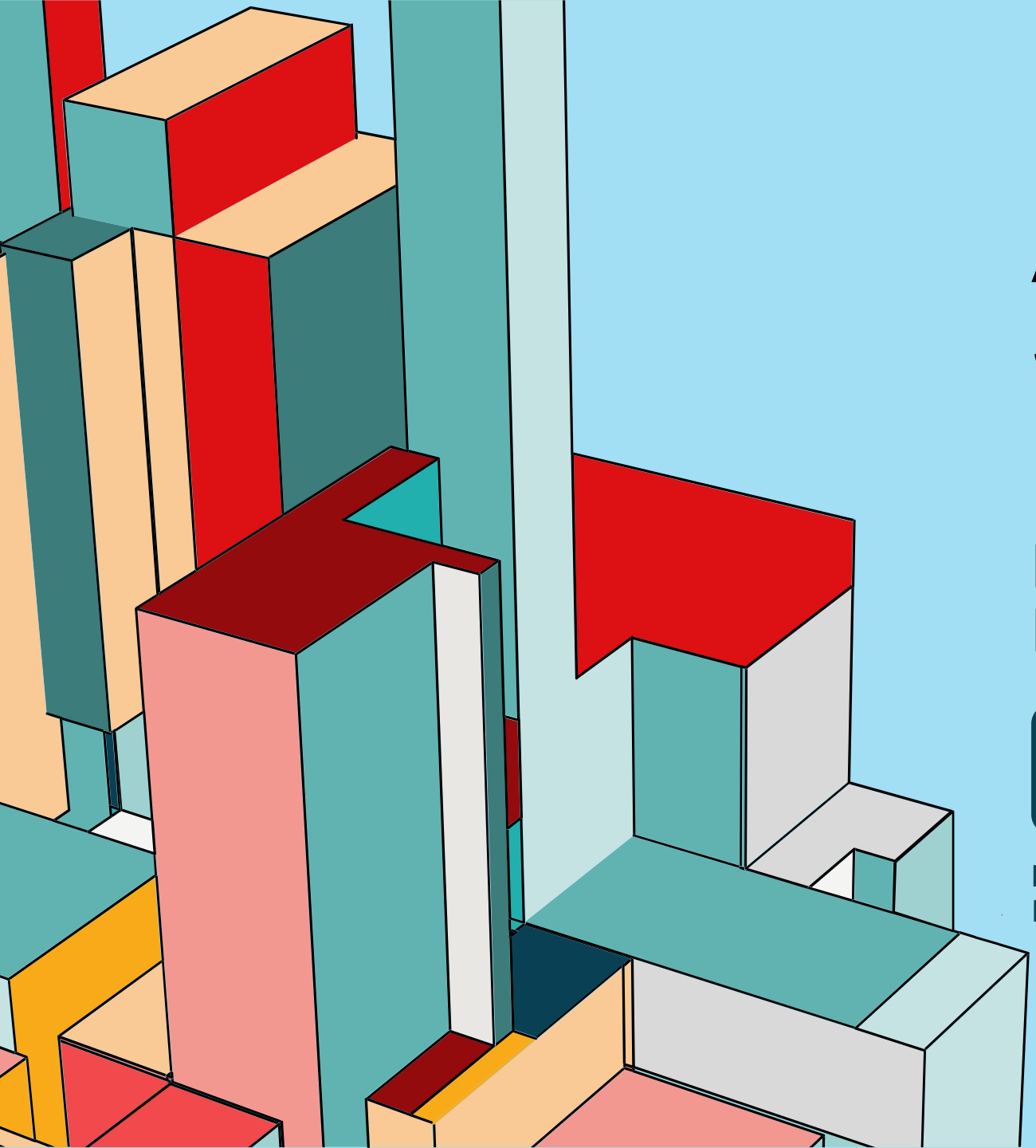


APPLIED DATA SCIENCE PROJECT

PREDICTING VLA FAILURES

Can we tell a robot is about to fail, before it does?

Proposed by **Lorenzo Natale**
Istituto Italiano di Tecnologia (IIT)

Politecnico di Torino

FONDAZIONE links PASSION FOR INNOVATION

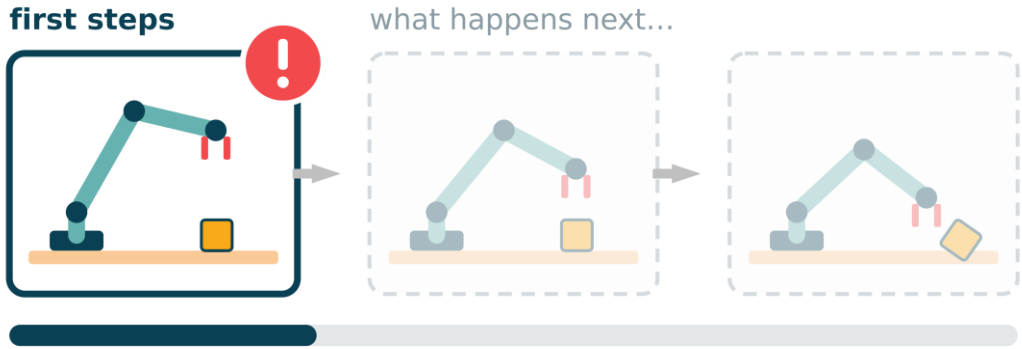
ellis European Laboratory for Learning and Intelligent Systems

VALUE-DRIVEN PROJECT

Spot the failure before it happens

WHY

Vision Language Action models follow many instructions but fail unpredictably. An early warning lets the robot stop, ask for help or retry.

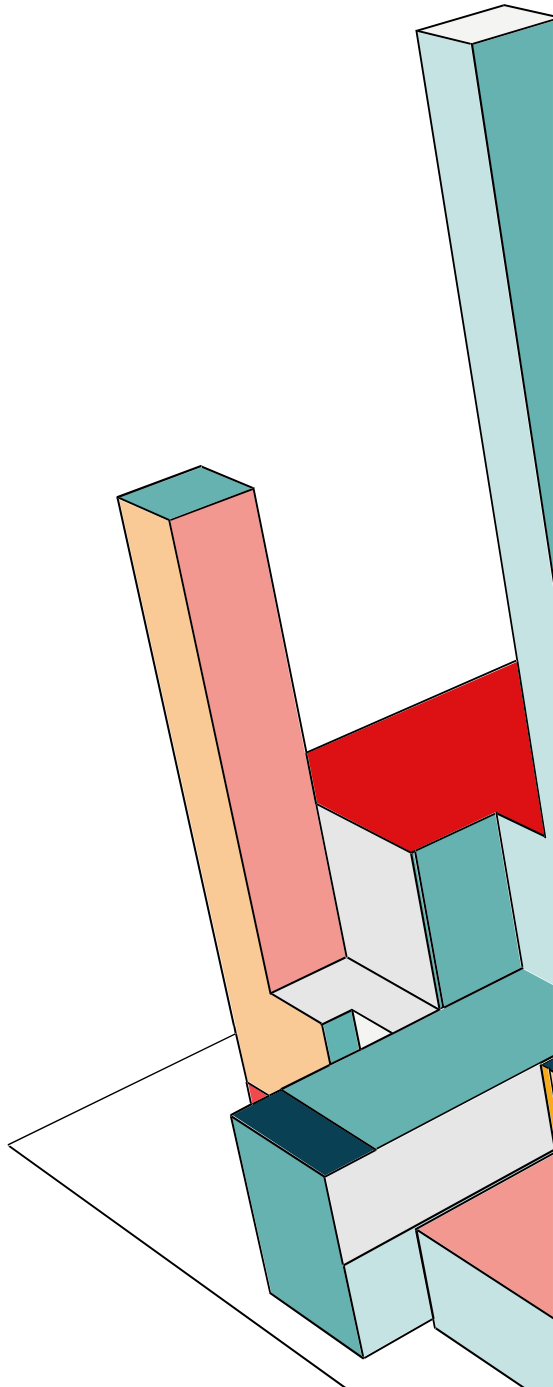


WHO BENEFITS

People who work or live alongside robots, in factories, warehouses, hospitals and homes.

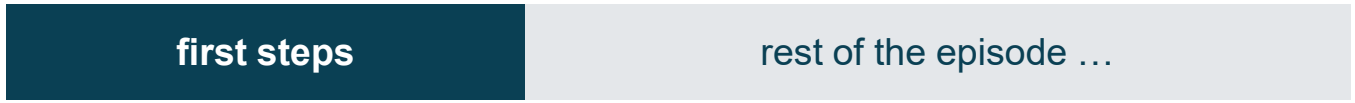
UN SDGs

SDG 9: Industry, Innovation and Infrastructure; indirectly SDG 8, through safer shared workplaces.

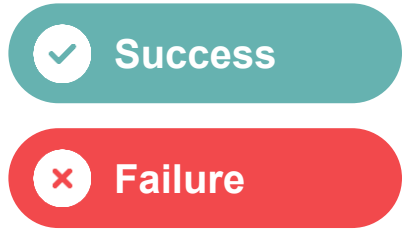


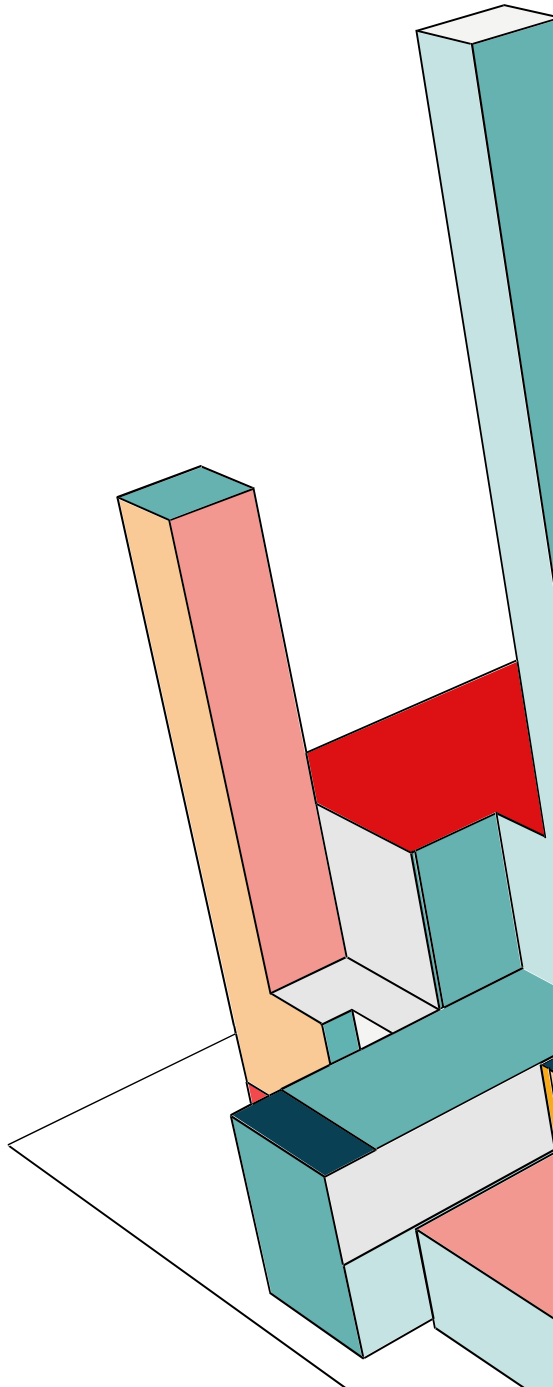
DATA

A dataset of successes and failures, generated in simulation



One episode of the VLA in simulation





01

Collect

Run a pretrained VLA such as GR00T N1.7 on LIBERO-10 tasks or in Isaac Sim; record images, robot states, actions and policy features.

02

Label

The simulator's task-completion check marks each episode. Vary tasks and starting conditions to get enough failures.

03

Check

Spot-check labels on videos, watch class balance, split train/test by episode, version scripts and seeds.

TASK

Watch the start, predict the end

Images

Robot states

Policy features

Failure classifier
binary classification

?

Will this episode fail?

METRICS

AUROC
and accuracy, on unseen episodes

Precision, recall
share of failures caught, and how reliably

False alarm rate
successes wrongly flagged as failures

Earliness
how early (as % of the episode) failures are predicted

Extension: a small world model that forecasts upcoming observations to anticipate failures further ahead.

